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School of Computer Science and Engineering

Department of Computer Science and Engineering

Multiple Language Translator Using Vosk API

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Outline

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Introduction

- In today’s globalized world, communication across different languages is a major challenge
- People from different regions often face difficulties in understanding each other due to language barriers
- Traditional translation methods are slow and require manual input
- With advancements in Artificial Intelligence, real-time speech recognition and translation have become possible
- These technologies enable seamless and instant communication between individuals speaking different languages

Introduction

- Speech recognition systems convert spoken language into text automatically
- When combined with translation systems, they provide real-time multilingual communication
- This project focuses on developing a system that integrates speech recognition and translation
- The aim is to create a **real-time, user-friendly multilingual translator**
- The system enhances accessibility, communication efficiency, and user convenience

What is Vosk?

- Vosk is an **offline speech recognition toolkit**
- Built on **Kaldi engine**
- Converts **speech into text in real-time**
- Works without internet connection
- Supports multiple programming languages like Java
- Uses **hybrid HMM + Deep Neural Network (DNN) models**

How Vosk Works

- Captures audio from microphone
- Extracts features using **MFCC (Mel Frequency Cepstral Coefficients)**
- Uses **HMM + DNN for speech recognition**
- Applies **language model** to form meaningful words
- Uses **Viterbi algorithm** for best sequence prediction
- Generates text in real-time

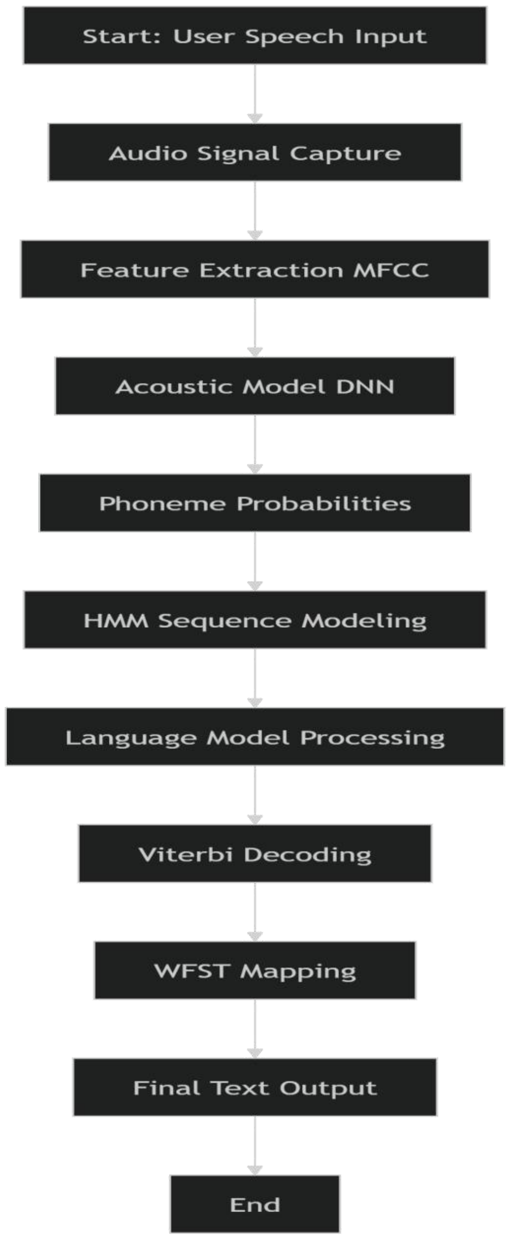
Features of Vosk

- Supports **20+ languages** (English, Hindi, Spanish, etc.)
- Works **offline → no internet required**
- Provides **real-time streaming recognition**
- Uses **pre-trained models** (small & large)
- Efficient and works on **low-resource systems**
- Ensures **privacy (local processing)**

Vosk Algorithm Working

•Audio signal is captured and converted into **feature vectors (MFCC)**
•**Acoustic model (DNN)** predicts probabilities of phonemes (basic speech units)
•**Hidden Markov Model (HMM)** models sequence of phonemes over time
•**Language model** predicts most meaningful word combinations
•**Viterbi algorithm** finds the most probable word sequence
•**WFST decoder** efficiently maps speech → text

“Vosk uses a hybrid model combining Hidden Markov Models (HMM) and Deep Neural Networks (DNN), with Viterbi decoding and WFST-based search.”



Literature Review

- Speech recognition systems have evolved significantly using AI and deep learning techniques
- Traditional systems were based on **Hidden Markov Models (HMM)**, while modern systems use **Deep Neural Networks (DNN)**
- Kaldi is widely used for building speech recognition systems
- Vosk is built on Kaldi and provides **offline real-time speech recognition**
- Research shows that hybrid HMM-DNN models improve accuracy in continuous speech recognition
- **References:**
- https://kaldi-asr.org
- https://alphacephei.com/vosk/

Literature Review

- Machine Translation has evolved from rule-based systems to **Neural Machine Translation (NMT)**
- Modern systems use **sequence-to-sequence models and transformers**
- APIs like **LibreTranslate** and Google Translate provide real-time translation services
- Research shows that NMT improves translation quality and contextual understanding
- Real-time translation systems combine speech recognition and machine translation for better usability
- **References:**
- https://libretranslate.com/
- https://ai.googleblog.com/2016/09/a-neural-network-for-machine.html

Literature Review

- Existing systems like Google Translate require **internet connectivity**
- Many systems lack **real-time speech-to-speech translation**
- Privacy concerns arise due to cloud-based processing
- Offline tools like Vosk address privacy and latency issues
- **Research Gap:** Need for a system that is:
    - Real-time
    - Offline-capable
    - Multi-language supported
    - Lightweight and efficient
- **References:**
- https://research.google/pubs/pub45610/
- https://alphacephei.com/vosk/

Problem Statement

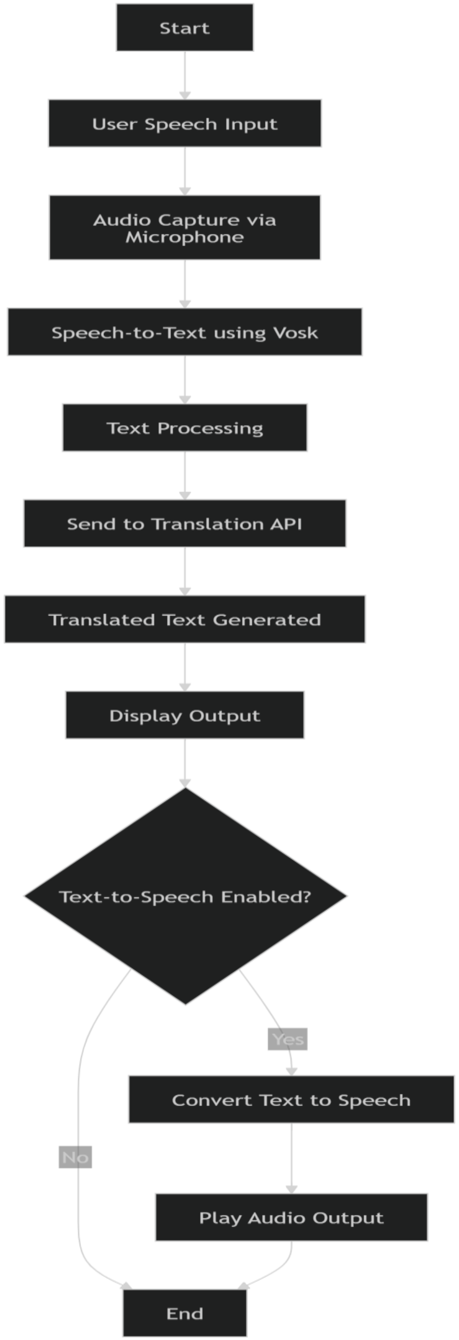
Existing translation systems are not efficient for **real-time speech-to-speech communication** and often require internet connectivity.

Objectives

- To develop a system that converts **speech into text in real-time**
- To translate the recognized text into **multiple languages efficiently**
- To provide **real-time output (text and optional speech)**
- To design a **user-friendly and offline-capable translation system**

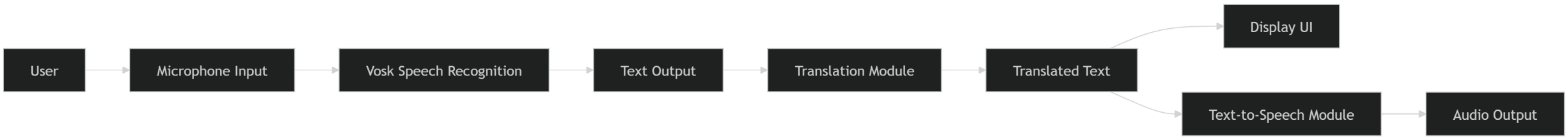
Proposed Methodology

•The system captures **speech input from the user**
•Audio is processed using **Vosk Speech Recognition**
•Speech is converted into **text format**
•The text is sent to the **translation module**
•Translated output is displayed to the user



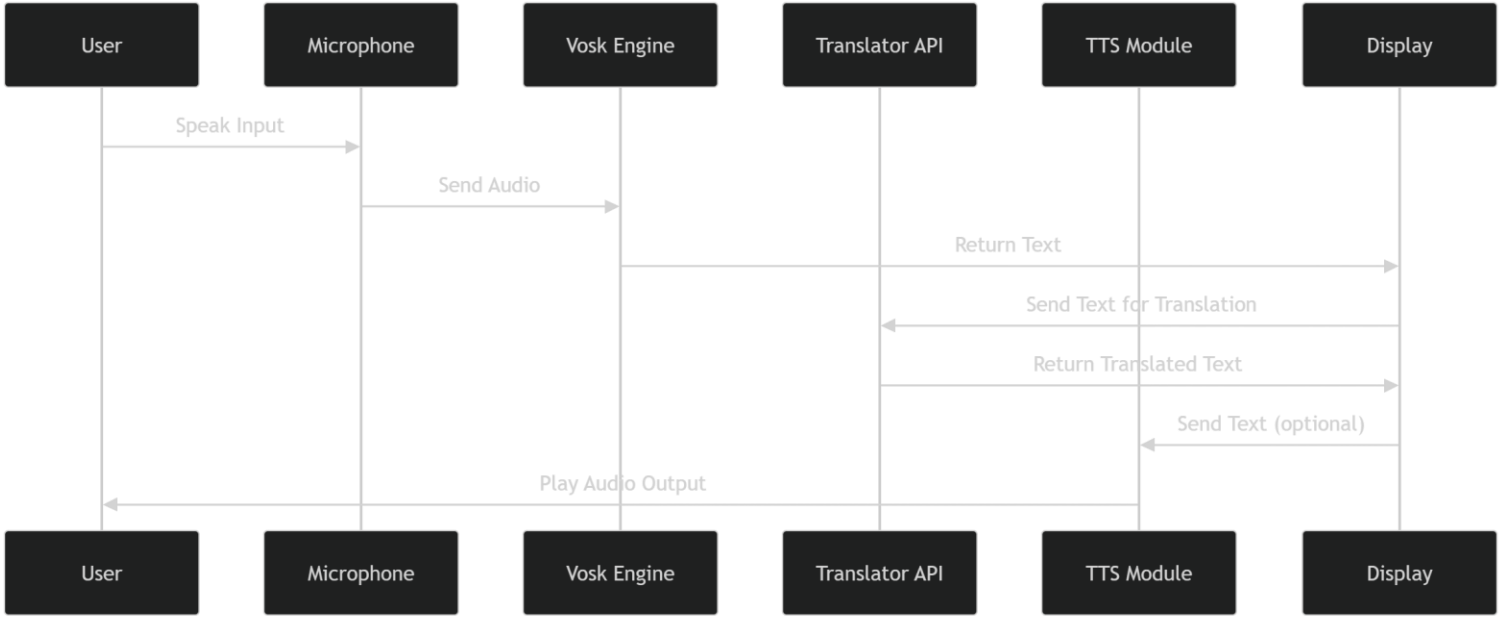
System Workflow

- User clicks **Start**
- Microphone captures speech
- Vosk converts speech → text
- Text is sent to translation API
- Translated text is generated
- Output is displayed (and optionally spoken)



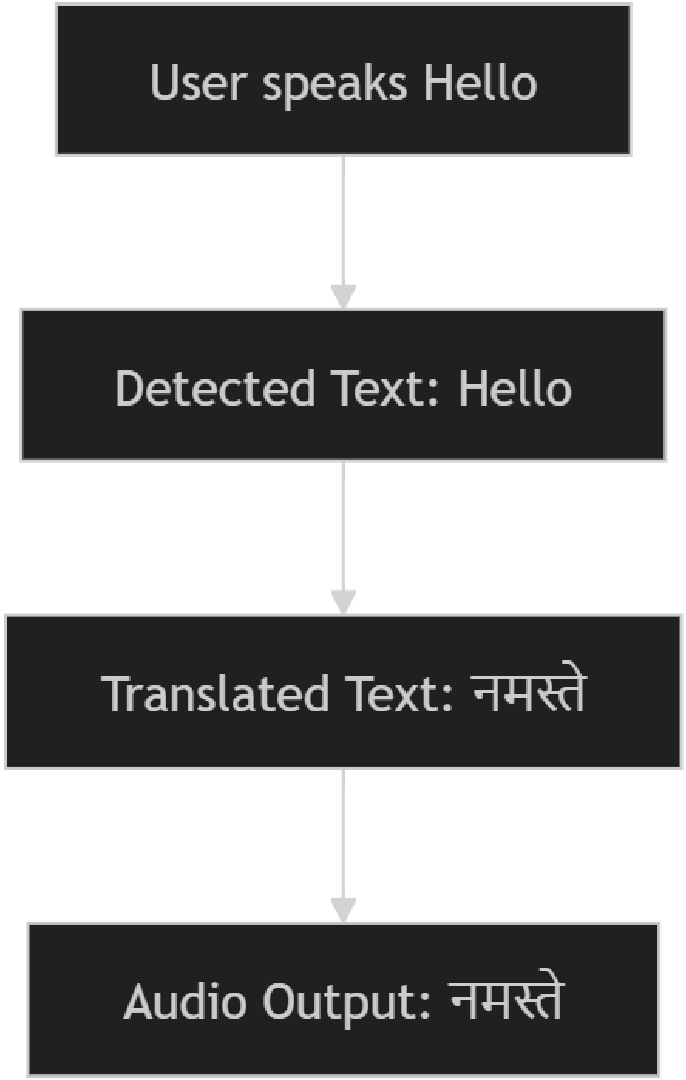
Module Interaction

- **Input Module:** Captures voice input
- **Speech Recognition Module:** Converts audio to text
- **Translation Module:** Converts text into target language
- **Output Module:** Displays and speaks translated result
- All modules work **sequentially in real-time**



Result

•System successfully converts **speech to text in real-time**
•Accurate translation into **multiple languages**
•Supports **real-time output with minimal delay**
•Optional **text-to-speech output works correctly**
•Performs efficiently in normal speaking conditions



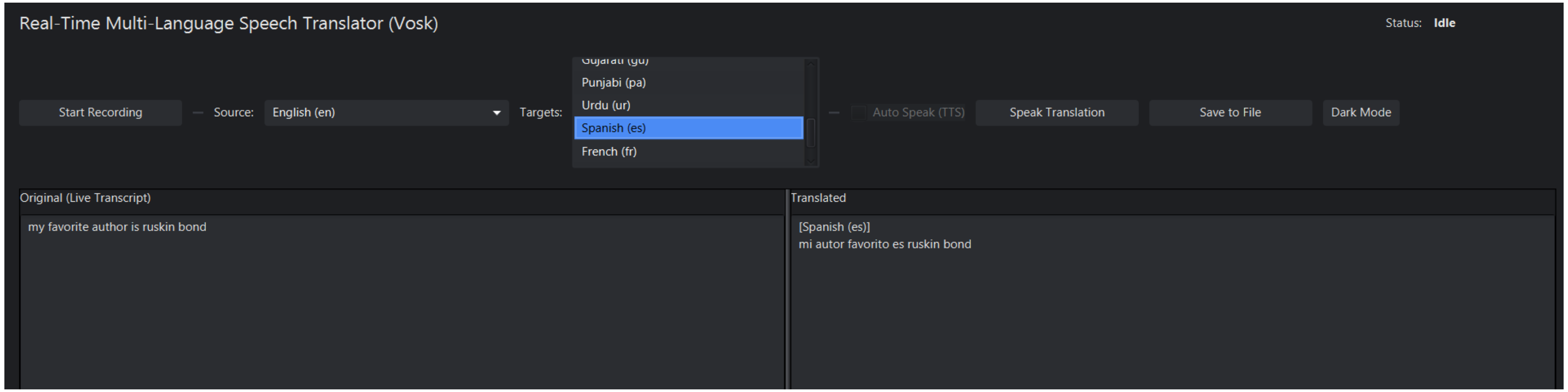
Speech Input → Text Output → Translated Text → Audio Output

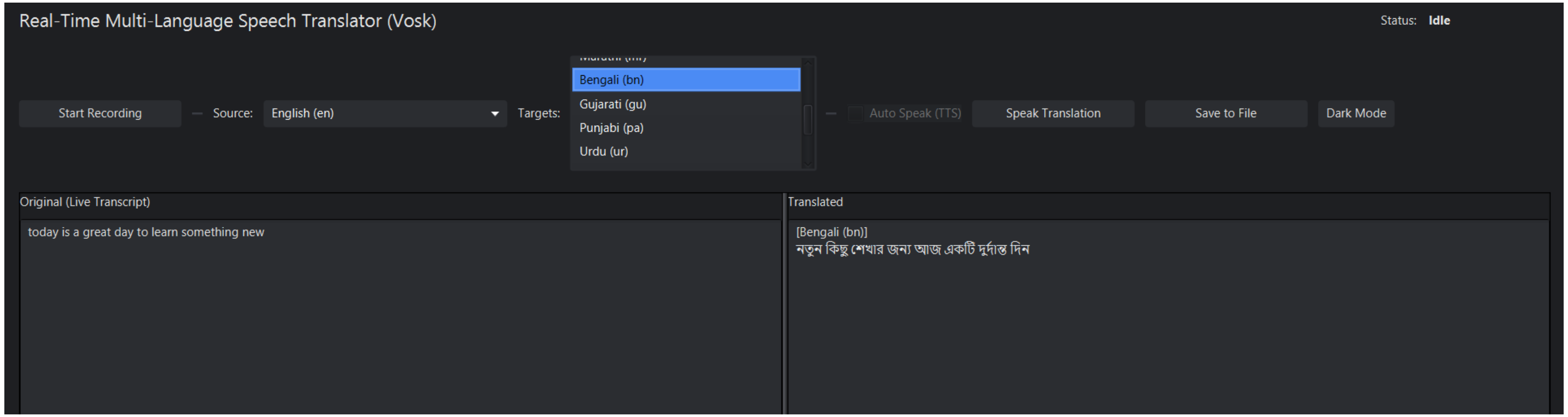
Example:

Input: *"Hello"*

Output: *"नमस्ते"* (Hindi)

Result


Real-Time Multi-Language Speech Translator (Vosk)
Status: Idle
Start Recording
Source:
English (en)
Targets:
Punjabi (pa)
Urdu (ur)
Spanish (es)
French (fr)
Auto Speak (TTS)
Speak Translation
Save to File
Dark Mode
Original (Live Transcript)
my favorite author is ruskin bond
Translated
[Spanish (es)]
mi autor favorito es ruskin bond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Bengali (bn)
Gujarati (gu)
Punjabi (pa)
Urdu (ur)
Auto Speak (TTS)
Speak Translation
Save to File
Dark Mode
Original (Live Transcript)
today is a great day to learn something new
Translated
[Bengali (bn)]
নতুন কিছু শেখার জন্য আজ একটি দুর্দান্ত দিন

Outcome

- Successfully developed a **real-time multilingual speech translator**
- Achieved accurate **speech-to-text conversion using Vosk**
- Implemented **multi-language translation with real-time output**
- Integrated **text-to-speech for audio output**
- System works efficiently with **low latency and good accuracy**

References

- Vosk API Documentation – https://alphacephei.com/vosk/
- Kaldi Speech Recognition Toolkit – https://kaldi-asr.org/
- LibreTranslate API – https://libretranslate.com/
- Java Documentation – https://docs.oracle.com/javase/
- Rabiner, L. (1989). *A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition*

Thank You